

Rian Fritz D. Jalandoni, MSc.

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Location: Montreal, Canada | Iloilo City, Philippines LinkedIn: LinkedIn GitHub: GitHub

EDUCATION

McGill University <i>PhD in Quantitative Life Science</i>	Montreal, Canada 2023 – Present
International Centre for Theoretical Physics <i>Postgraduate Diploma in Quantitative Life Sciences</i> • Thesis: <i>Dynamics of systems of replicators with cooperation</i> [https://ictp.primo.exlibrisgroup.com/permalink/39ICT_INST/1bcnctb/alma990001607490109086]	Trieste, Italy 2022 – 2023
University of the Philippines-Diliman <i>MSc in Physics</i> • Grade: 1.03 (Outstanding Graduate Student) • Thesis: <i>Numerical Analysis of Cancer Growth on a Spherical Surface and 3D Continuous Space</i>	Philippines 2020 – 2022
University of the Philippines-Diliman <i>BSc in Applied Physics, Major in Instrumentation</i> • Grade: 1.4546 (cum laude) • Thesis: <i>Mechanical Model of Cancer Cells in 2D Confined Continuous Space</i>	Philippines 2015 – 2020
Philippine Science High School – Western Visayas Campus <i>High School</i>	Philippines 2011 – 2015

PROFESSIONAL EXPERIENCE

Student Researcher <i>Instrumentation Physics Laboratory – Complexity Science Group</i> • National Institute of Physics, University of the Philippines-Diliman • Conducted research on physical and mechanical effects of cancer growth in 3D volume. • Developed simulation models to analyze mechanical stress in spherical cancerous tissue using Voronoi Tessellations.	October 2017 – July 2022
Summer School Research Student <i>London Mathematical Laboratory School</i> • Project: <i>Stability of ecosystems with a core-periphery structure</i> • Participated in advanced research projects focusing on mathematical modeling in ecological systems.	July 2021
Bioinformatics Intern <i>Philippine Genome Center, University of the Philippines-Diliman</i> • Project: <i>A directed minimum connected dominating set for protein-protein interaction networks</i> • Applied computational methods to analyze genomic data and developed algorithms for biological network analysis.	June 2019 – July 2019

RESEARCH INTERESTS

Areas of Interest: Biological Modelling and Biophysics, Learning and Information Processing, Network Algorithms and Analysis, NeuroAI, Computational Neuroscience and Psychiatry, Ecology and Evolution, Numerical Simulations in Biology, Statistical Physics in Biological Systems

Areas of Research Biological Modelling, Dynamical Systems in Biology, Computational Neuroscience, Behavioural Modelling, NeuroAI

PUBLICATIONS AND PRESENTATIONS

Qiu, H., Jalandoni, R.F., Derham, A., Reinisch, N., Chen, C., Vong, A., Cook, E.P., Krishnaswamy, A., Khadra, A. and Sharif-Naeini, R., 2026. Astrocytic FMRP regulates the function of spinal parvalbumin-expressing neurons in Fragile X Syndrome. *bioRxiv*, pp.2026-07.

RFD Jalandoni and JY Bantang, “Physical and mechanical effects of cancer growth in 3D volume,” *Proceedings of the Samahang Pisika ng Pilipinas* 40, SPP-2022-2F-06 (2022).
<https://proceedings.spp-online.org/article/view/SPP-2022-2F-06>

R. Jalandoni and J. Bantang, “Simulation of Nutrient-limited Cancer Growth on a Spherical Surface,” DOST-ASTHRDP Scholars Graduate Scholars Conference 2022, Sept. 22–23, 2022. (Abstract and Poster Presentation)

RFD Jalandoni and JY Bantang, “Numerical analysis of mechanical stress in a spherical cancerous tissue,” *Proceedings of the Samahang Pisika ng Pilipinas* 39, SPP-2021-2G-05 (2021).
<https://proceedings.spp-online.org/article/view/SPP-2021-2G-05>

R. Ramos, R. Jalandoni, J. Bantang, “Cellular Automaton-based Model of Neuronal Dynamics on a Spherical Surface,” International Workshop on Mathematical Biology 2021. (Abstract and Poster Presentation)

RFD Jalandoni and JY Bantang, “In silico study of stress distribution in a tissue surrounding a cancer growth using Centroidal Voronoi Tessellations,” *Proceedings of the Samahang Pisika ng Pilipinas* 38, SPP-2020-4A-03 (2020).
<https://paperview.spp-online.org/proceedings/article/view/SPP-2020-4A-03>

Briones, Y. L., Castro, M. R., Jalandoni, R. D., Adorna, H. N., Dizon, J. A., Young, A. T., “A directed minimum connected dominating set for protein-protein interaction networks,” in *Theory and Practice of Computation*, pp. 176–187, CRC Press (2020).

R.F. Jalandoni, J. Bantang, “Mechanical stress properties by a 2D model of cancer growth using Centroidal Voronoi Tessellations,” BrainsConnect 2019, Oct. 23–24, 2019, Henry Sy Sr. Auditorium, St. Luke’s Medical Center-Global City, Taguig City, Philippines.

R.F. Jalandoni, J. Bantang, “Mechanical stress generated by a 2D model of Cancer growth using a Centroidal Voronoi Tessellations,” 16th International Conference on Molecular Systems Biology (ICMSB 2019), Oct. 28–31, 2019, De La Salle University, Manila, Philippines. (Abstract and Poster Presentation)

CONFERENCES AND TALKS

<i>International Conference on Mathematical Neuroscience–Poster</i>	<i>June 2026</i>
<i>Center for Applied Mathematics in Bioscience and Medicine Workshop–Short Presentation</i>	<i>May 2026</i>
<i>19th Canadian Neuroscience Meeting–Attendee</i>	<i>May 2026</i>
<i>McGill Quantitative Life Science Research Day–Poster</i>	<i>June 2025</i>
<i>Center for Applied Mathematics in Bioscience and Medicine Workshop–Short Presentation</i>	<i>February 2023</i>
<i>40th Samahang Pisika ng Pilipinas Physics Conference – Parallel Presentation Presenter</i>	<i>October 2022</i>
<i>DOST-ASTHRDP Graduate Scholars’ Conference 2022 – Participant and Poster Presenter</i>	<i>September 2022</i>
<i>Frontiers in Theoretical Biophysics – Participant</i>	<i>November 2021</i>
<i>39th Samahang Pisika ng Pilipinas Physics Conference – Parallel Presentation Presenter</i>	<i>October 2021</i>
<i>International Workshop on Mathematical Biology (IWOMB) 2021 – Poster Presenter</i>	<i>June 2021</i>
• Award: 3rd Best Poster Presenter	
<i>6th Oxford Intl. Neuron and Brain Mechanics Workshop – Participant</i>	<i>April 2021</i>
<i>ICTP Conference on Quantum Annealing/Adiabatic Quantum Computation – Participant</i>	<i>October 2020</i>
<i>38th Samahang Pisika ng Pilipinas Physics Conference – Short Presentation Presenter</i>	<i>October 2020</i>
<i>16th International Conference on Molecular Systems Biology (ICMSB 2019) – Poster Presenter</i>	<i>October 2019</i>
<i>BrainsConnect 2019 – 9th Neuroscience International Symposium – Poster Presenter</i>	<i>October 2019</i>

PROFESSIONAL DEVELOPMENT ACTIVITIES

<i>ICTP Spring College on the Physics of Complex Systems</i>	<i>February – March 2023</i>
• International Centre for Theoretical Physics	

AFFILIATIONS

<i>Samahang Pisika ng Pilipinas (Physics Society of the Philippines) – Associate Member</i>	<i>2020 – Present</i>
<i>Instrumentation Physics Laboratory, NIP, UP Diliman – Member / Grad. Student Researcher</i>	<i>2017 – 2022</i>
<i>UP Kasanag (UP Regional Organization for Western Visayas) – Publicity Committee Member</i>	<i>2018 – 2022</i>

TECHNICAL SKILLS

Programming Languages: Python, Julia, C, C++, PHP, HTML, Matlab

Models & Methods: TensorFlow, Qiskit, Neural Networks, NLP, Image Processing, ARIMA Models

Tools: GIMP, ImageJ, Minitab, Adobe Photoshop, Microsoft Office, LaTeX